

Bank Loan Analysis Project Report

A comprehensive data analytics project focused on evaluating loan applications, funded amounts, repayment behavior, customer segmentation, and business risk patterns using SQL Server, and Power BI.

1. Project Overview

This project analyzes financial loan application data to uncover lending trends, borrower behavior, repayment performance, and portfolio quality. The objective is to help financial institutions monitor loan performance, reduce bad debt risk, optimize approval strategies, and improve decision-making through data-driven insights.

2. Dataset Summary

- Dataset: financial_loan.csv
- Includes customer demographics, employment details, loan purpose, funded amount, interest rates, debt-to-income ratio, repayment status, and regional data.
- Key dimensions: Loan Status, Term, Purpose, Home Ownership, Employment Length, State, Issue Date
- Key KPIs: Total Applications, Funded Amount, Amount Received, Good Loan %, Bad Loan %, Interest Rate, DTI Ratio

4. SQL Business Analysis

Using SQL Server, multiple KPI dashboards and business intelligence queries were developed:

- Total Loan Applications (Overall, MTD, PMTD)

	Total_NumberOf_Application
1	38576

	MTD_Total_Loan_Application
1	4314

	PMTD_Total_Loan_Application
1	4035

- Total Funded Amount and Total Amount Received

	Total_Funded_Amount
1	435757075

	MTD_Total_Funded_Amount
1	53981425

	PMTD_Total_Funded_Amount
1	47754825

	Total_Amount_Received
1	473070933

	MTD_Total_Amount_Received
1	58074380

	PMTD_Total_Amount_Received
1	50132030

- Average Interest Rate and Debt-to-Income Ratio

	Average_interast_Rate
1	12.04883

	MTD_Average_interast_Rate
1	12.35604

	PMTD_Average_interast_Rate
1	11.94172

	Average_Debt_to_Income_Ratio
1	13.32743

	MTD_Average_Debt_to_Income_Ratio
1	13.66554

	PMTD_Average_Debt_to_Income_Ratio
1	13.30273

- Good Loan vs Bad Loan Performance

	Good_Loan_Percentage
1	86

	Bad_Loan_Percentage
1	13

- Loan Status Analysis

	loan_status	Total_loan_Application	Average_DTI	Average_Interest_Rate	Total_Funded_Amount	Total_Received_Amount
1	Fully Paid	32145	13.16735	11.64107	351358350	411586256
2	Charged Off	5333	14.00473	13.87857	65532225	37284763
3	Current	1098	14.72434	15.09933	18866500	24199914

- Monthly Trend Analysis

	Month_NO	Month_Name	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

- Regional (State-wise) Lending Analysis

	address_state	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660

- Employment Length Analysis

	emp_length	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	10+ years	8870	116115950	125871616
2	< 1 year	4575	44210625	47545011
3	2 years	4382	44967975	49206961
4	3 years	4088	43937850	47551832
5	4 years	3428	37600375	40964850
6	5 years	3273	36973625	40397571
7	1 year	3229	32883125	35498348
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

- Loan Purpose Analysis

	purpose	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	Debt consolidation	18214	232459675	253801871
2	credit card	4998	58885175	65214084
3	other	3824	31155750	33289676
4	home improvement	2876	33350775	36380930
5	major purchase	2110	17251600	18676927
6	small business	1776	24123100	23814817
7	car	1497	10223575	11324914
8	wedding	928	9225800	10266856
9	medical	667	5533225	5851372
10	moving	559	3748125	3999899
11	house	366	4824925	5185538
12	vacation	352	1967950	2116738
13	educational	315	2161650	2248380
14	renewable_energy	94	845750	898931

- Home Ownership Analysis

	home_ownership	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	RENT	18439	185768475	201823056
2	MORTGAGE	17198	219329150	238474438
3	OWN	2838	29597675	31729129
4	OTHER	98	1044975	1025257
5	NONE	3	16800	19053

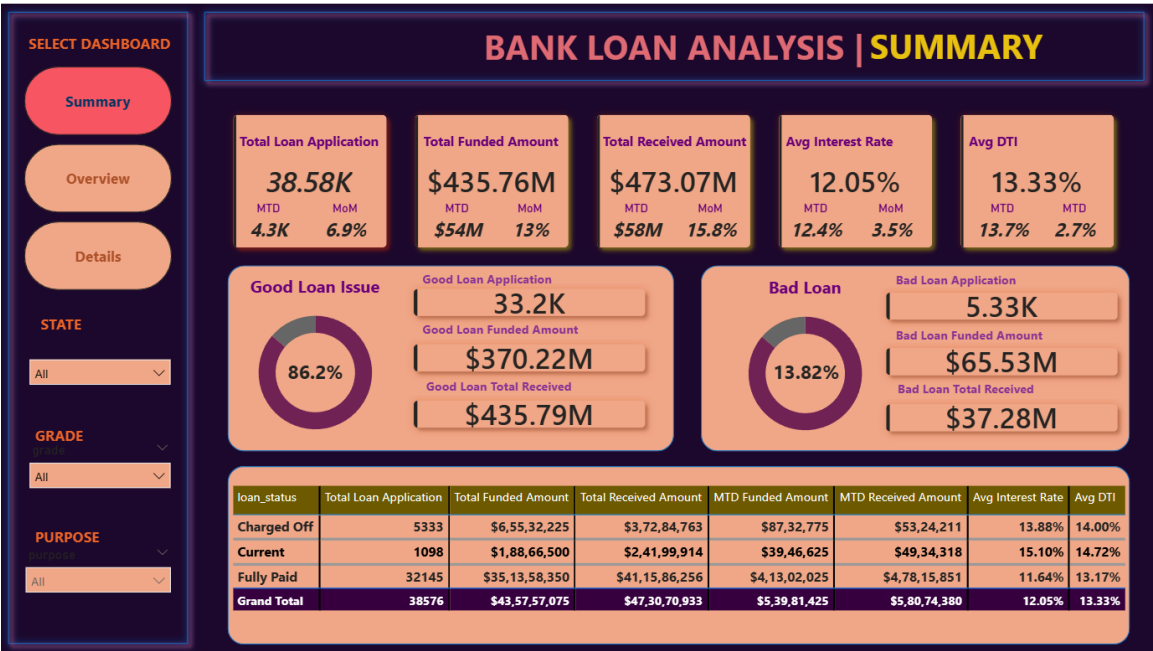
5. Key Business Insights

- Good loans (Fully Paid + Current) represent the majority of applications and funded capital.
- Bad loans (Charged Off) highlight financial risk exposure requiring stricter approval strategies.
- Debt consolidation is the most common loan purpose category.
- Mortgage and Rent borrowers form the largest home ownership segments.
- Borrowers with longer employment history generally contribute more stable repayment behavior.
- Monthly application and funding trends help identify seasonal lending patterns.

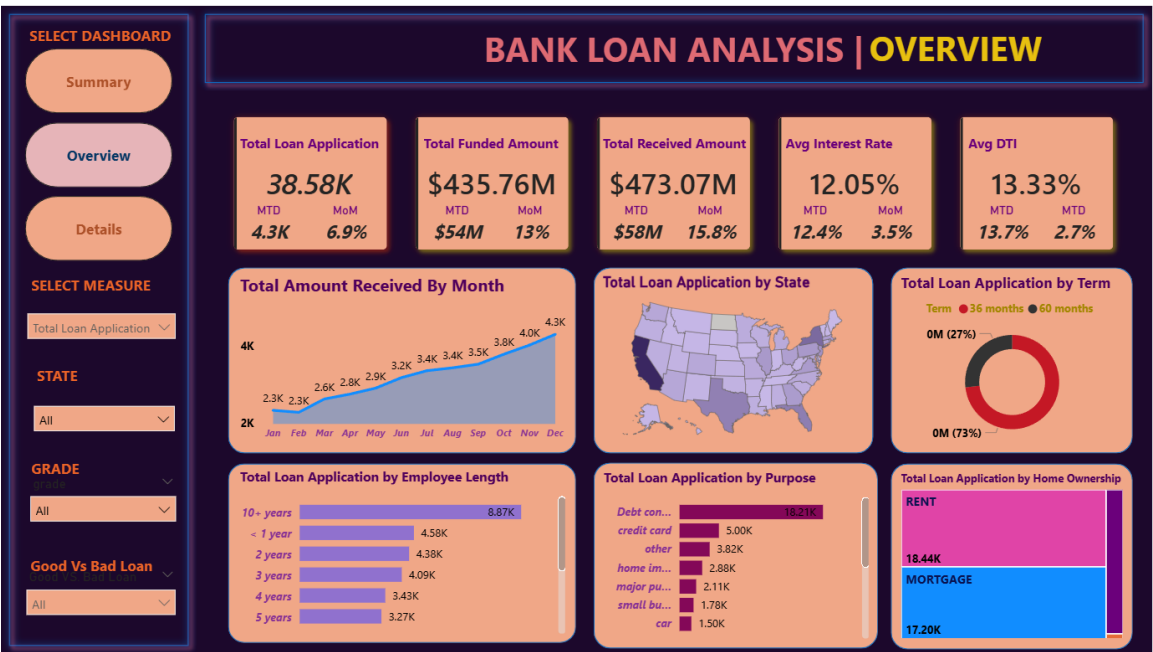
6. Power BI Dashboard

An interactive Power BI dashboard was developed to visualize:

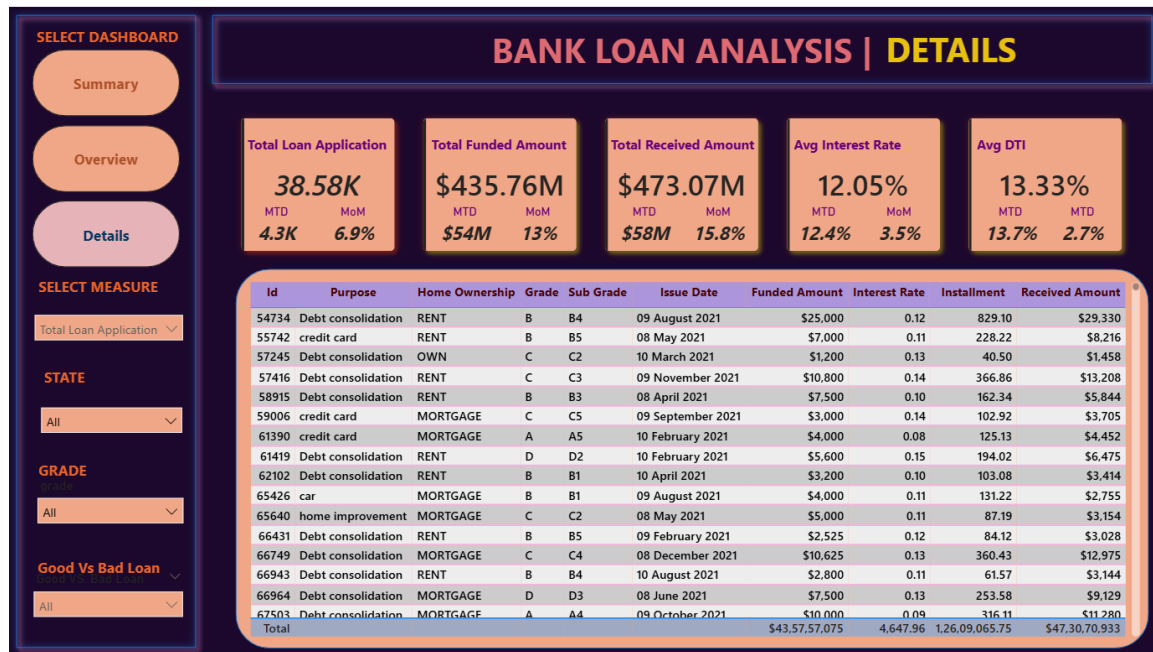
SUMMARY DASHBOARD:



OVERVIEW DASHBOARD:



DETAILS DASHBOARD:



7. Tools & Technologies Used

- SQL Server for business query analysis
- Power BI for dashboard creation
- Excel data source

8. Business Recommendations

1. Reduce Bad Loan Risk

Tighten approval criteria for high-DTI, low-credit-grade, and unstable employment borrowers. Use predictive risk models to lower charge-offs.

2. Expand Good Loan Segments

Focus more on reliable borrowers such as long-term employees, mortgage holders, and high-grade applicants. Offer faster approvals and loyalty benefits.

3. Strengthen Debt Consolidation Products

Since debt consolidation has the highest application volume, create competitive products in this segment. Add repayment support programs to reduce defaults.

4. Optimize Regional & Seasonal Lending

Invest more in high-performing states and peak lending months (especially year-end).

Adjust marketing and operational resources based on demand trends.

5. Improve Collection & Recovery Systems

Build early warning systems for risky accounts.

Use automated reminders, restructuring options, and proactive collections to improve recovery rates.